

Abstract

The Cayley–Dickson octonion algebra has two layers of structure relevant to compiler optimisation. *Categorically*, its reassociation law is a cocycle-coherent, everywhere-invertible associator: the sign factor $\varphi \in \{+1, -1\}$ satisfies the 3-cocycle condition of Albuquerque–Majid and thus the pentagon axiom of Lack–Street’s skew monoidal framework. *Operationally*, only a Fano-selected subset of these reassociations are semantics-preserving compiler rewrites: exactly 175 of 343 ordered triples of imaginary basis elements yield identical results under both parenthesisations (42 from Fano-line collinearity in $\text{PG}(2,2)$ and 133 from repeated-index alternativity), while the remaining 168 triples change sign, with $168 = |\text{PSL}(2,7)|$. We formalise this two-layer structure and implement it in Sounio, a self-hosted programming language. The type checker requires the `NonAssoc` algebraic effect for non-associative multiplication; the e-graph optimiser uses the Fano predicate to decide which reassociations are safe. The 168 sign-changing triples decompose under a dagger involution into $84 + 84$ forward/backward pairs with zero fixed points. All results are formalised in Lean 4 (122 theorems, no `sorry`, no Mathlib dependency) and cross-validated against the running compiler.

Keywords: Formal verification · Lean 4 · Automated reasoning · Decidability · Non-associative algebra · Fano plane · Cayley–Dickson · Skew monoidal category · Compiler optimisation · Algebraic effects

1 Introduction

The associator of an algebra measures the failure of associativity:

$$[a, b, c] = (a \cdot b) \cdot c - a \cdot (b \cdot c).$$

In an associative algebra the associator vanishes identically, and a compiler may freely rewrite $(a \cdot b) \cdot c$ as $a \cdot (b \cdot c)$. In a non-associative algebra it does not, and the compiler must preserve the programmer’s chosen parenthesisation.

But in the Cayley–Dickson algebras—the tower $\mathbb{R} \rightarrow \mathbb{C} \rightarrow \mathbb{H} \rightarrow \mathbb{O} \rightarrow \mathbb{S} \rightarrow \dots$ that extends the real numbers through complex numbers, quaternions, and octonions—the situation is more subtle. Not all triples are non-associative. For the octonions ($k = 3$), exactly 175 of 343 ordered triples of imaginary basis elements are associative, and 168 are not. Among the 210 triples with three distinct indices, the associative ones are precisely those whose indices form a line of the *Fano plane* $\text{PG}(2,2)$: $[e_i, e_j, e_l] = 0$ for distinct nonzero i, j, l iff $\{i, j, l\}$ is one of the 7 lines of $\text{PG}(2,2)$ [5]. The remaining 133 triples (with a repeated index) are associative by the alternativity of the octonions.

This paper introduces *Fano-selective reassociation*, a two-layer structure for non-associative type systems. Categorically, the octonion reassociation is cocycle-coherent and invertible: the sign factor $\varphi(a, b, c) \in \{+1, -1\}$ satisfies the 3-cocycle condition of Albuquerque–Majid [4] and thus the pentagon axiom of the skew monoidal framework [21, 12] (where associators are not required to be isomorphisms, though they may be). Operationally, only the 175 reassociations with $\varphi = +1$ are semantics-preserving compiler rewrites; the 168 with $\varphi = -1$ change the sign of the computed value. We define a *rewrite category* whose morphisms are the sign-preserving reassociations, and show that the Fano plane determines which morphisms exist.

Contributions.

1. **Two-layer structure** (Definition 3.1, Theorem 3.7): a cocycle-coherent associator at the algebraic level, and a Fano-selective rewrite discipline at the compiler level, with the Fano plane of $\text{PG}(2,2)$ determining which rewrites are semantics-preserving (§3).

2. **The 168 Theorem, formalised** (§5): 122 theorems in Lean 4, including the exact counts $175 + 168 = 343$, the binary norm property, and the 84/84 dagger decomposition. No sorry. No Mathlib dependency. Build time under 3 seconds.
3. **The NonAssoc algebraic effect** (§4): an implementation in Sounio where the type checker requires the `NonAssoc` effect for multiplication in non-associative algebras ($k \geq 3$), and the optimiser uses the ternary Fano predicate to decide reassociation safety for specific basis triples. This makes $\text{PG}(2, 2)$ an operational part of the compiler.
4. **Decidability methodology** (§5.1): a Lean 4 formalisation strategy based on exhaustive enumeration over finite domains and `native.decide`, requiring no Mathlib and building in under 3 seconds. The approach extends directly to higher Cayley–Dickson levels by increasing the enumeration range.

2 Background

2.1 Cayley–Dickson Algebras and the Sign Function

The Cayley–Dickson construction [9] builds a tower of 2^k -dimensional algebras:

$$\mathbb{R} \xrightarrow{k=1} \mathbb{C} \xrightarrow{k=2} \mathbb{H} \xrightarrow{k=3} \mathbb{O} \xrightarrow{k=4} \mathbb{S} \rightarrow \dots$$

At each step, commutativity or associativity is lost. The multiplication of basis elements is determined by a recursive *sign function* $\sigma : \{0, \dots, 2^k - 1\}^2 \times \mathbb{N} \rightarrow \{+1, -1\}$:

$$e_a \cdot e_b = \sigma(a, b, k) \cdot e_{a \oplus b}$$

where $\oplus$ denotes bitwise XOR. The sign function satisfies the recursive Cayley–Dickson doubling formula (see Ren & Zhao [18] for the explicit twist function).

The *associator* of basis elements (e_i, e_j, e_l) decomposes as:

$$[e_i, e_j, e_l] = (\alpha - \beta) \cdot e_{i \oplus j \oplus l} \tag{1}$$

where $\alpha = \sigma(i, j) \sigma(i \oplus j, l)$ and $\beta = \sigma(j, l) \sigma(i, j \oplus l)$. Since $\alpha, \beta \in \{+1, -1\}$, we have $\alpha - \beta \in \{-2, 0, +2\}$. We write $\text{wave}(i, j, l) := \alpha - \beta$ for the *associator coefficient* of the triple.

Theorem 2.1 (Two-Valued Associator). *For all $k \geq 3$ and all nonzero $i, j, l \in \{1, \dots, 2^k - 1\}$: the associator coefficient $\alpha - \beta \in \{-2, 0, +2\}$, so $\| [e_i, e_j, e_l] \| \in \{0, 2\}$.*

For $k = 3$ (octonions), the $7^3 = 343$ ordered triples of imaginary basis elements partition as follows:

Class	Count	Condition
Fano-line (distinct, collinear)	42	i, j, l distinct, collinear in $\text{PG}(2, 2)$
Repeated-index (alternativity)	133	$i=j$ or $j=l$ or $i=l$
<i>Associative subtotal</i>	175	$\alpha = \beta$
Forward (non-associative)	84	$\alpha - \beta = +2$
Backward (non-associative)	84	$\alpha - \beta = -2$
<i>Non-associative subtotal</i>	168	$\alpha \neq \beta$
Total	343	

The number $168 = |\text{PSL}(2, 7)|$ is the order of the automorphism group of the Fano plane $\text{PG}(2, 2)$ [5].

2.2 Skew Monoidal Categories

A *skew monoidal category* [21, 12] is a category $\mathcal{C}$ equipped with a bifunctor $\otimes : \mathcal{C} \times \mathcal{C} \rightarrow \mathcal{C}$, a unit object I , and natural transformations

$$\begin{aligned}\alpha_{A,B,C} &: (A \otimes B) \otimes C \rightarrow A \otimes (B \otimes C) \\ \lambda_A &: I \otimes A \rightarrow A \\ \rho_A &: A \rightarrow A \otimes I\end{aligned}$$

that are *not* required to be isomorphisms, subject to five coherence conditions. When all structure maps are invertible, this recovers an ordinary monoidal category.

The key insight: in a skew monoidal category, re-parenthesisation is a *directed* operation. The Tamari lattice [22] on binary trees makes this precise: right-rotation $((a \cdot b) \cdot c) \mapsto (a \cdot (b \cdot c))$ is a valid step, but the reverse is not.

2.3 Dagger Categories

A *dagger category* [20] is a category $\mathcal{C}$ with an involutive, identity-on-objects, contravariant functor $\dagger : \mathcal{C}^{\text{op}} \rightarrow \mathcal{C}$. For every morphism $f : A \rightarrow B$, there exists $f^\dagger : B \rightarrow A$ with $(f^\dagger)^\dagger = f$.

In categorical quantum mechanics [1], the dagger models time-reversal: reflecting a string diagram about a horizontal axis gives the dagger of the process.

2.4 Algebraic Effects

Algebraic effects [17] provide a modular account of computational side effects. A function's type signature declares which effects it may perform:

$$f : \tau_1 \rightarrow \tau_2 \text{ with } \rho$$

where ρ is an *effect row*—a set of named effects. Effect rows form a Boolean algebra under union, intersection, and complement, with subrow inclusion providing effect subtyping [14, 15].

3 Fano-Selective Reassociation: Two Layers

Definition 3.1 (Fano-selective reassociation structure). Let A be a 2^k -dimensional Cayley–Dickson algebra with basis $\{e_0, \dots, e_{2^k-1}\}$, product $e_i \cdot e_j = \sigma(i, j, k) \cdot e_{i \oplus j}$. Let $B_0 = \{0, \dots, 2^k - 1\}$ index all basis elements and $B = \{1, \dots, 2^k - 1\}$ the imaginary ones. A *Fano-selective reassociation structure* on A consists of two layers:

Algebraic layer (cocycle).

1. A reassociation factor $\varphi : B_0^3 \rightarrow \{+1, -1\}$ defined by $\varphi(i, j, l) = \alpha(i, j, l) \cdot \beta(i, j, l)$, where α and β are the left and right parenthesisatation signs. This factor is a normalised 3-cocycle on $\mathbb{Z}_2^k$ (§3.4).
2. For every triple $(i, j, l) \in B_0^3$, a signed reassociation map

$$\alpha_{Ri,j,l} : (e_i \cdot e_j) \cdot e_l \mapsto \varphi(i, j, l) \cdot e_i \cdot (e_j \cdot e_l)$$

that is always invertible as a linear map (since $\varphi^{-1} = \varphi$).

Rewrite layer (compiler).

3. A decidable predicate $\mathcal{F} : B^3 \rightarrow \{0, 1\}$, defined by $\mathcal{F}(i, j, l) = 1$ iff $\varphi(i, j, l) = +1$.
4. For *associative* triples ($\mathcal{F}(i, j, l) = 1$), the two parenthesisations yield identical results, so the reassociation is a valid semantics-preserving compiler rewrite.
5. For *sign-changing* triples ($\mathcal{F}(i, j, l) = 0$), the reassociation changes the computed value by a factor of -1 , so no semantics-preserving rewrite exists. These triples have no morphism in the *rewrite category* (§3.5).

A *dagger* on such a structure is an involution $\dagger : B^3 \rightarrow B^3$, $(i, j, l) \mapsto (l, j, i)$, such that:

- The associative set is dagger-closed: $\mathcal{F}(i, j, l) = 1 \implies \mathcal{F}(l, j, i) = 1$.
- The dagger negates the associator coefficient on sign-changing triples: $\mathcal{F}(i, j, l) = 0 \implies \text{wave}(l, j, i) = -\text{wave}(i, j, l)$.

Remark 3.2. The algebraic layer is an everywhere-invertible cocycle associator in the sense of Albuquerque–Majid [4]. The rewrite layer extracts from it a selective discipline: the compiler treats sign-preservation as an effect boundary, blocking reassociations that change the computed value. These are genuinely two different categories (§3.4, §3.5).

Remark 3.3. When $\mathcal{F} \equiv 1$ (all triples), every reassociation is valid and the algebra is associative. When $\mathcal{F} \equiv 0$, no reassociation is safe. The Fano-selective structure interpolates between these extremes. The relationship to the partially normal skew monoidal categories of Uustalu–Veltri–Zeilberger [23], where selected structural laws are invertible, deserves further investigation.

3.1 The Octonionic Instance

For the Cayley–Dickson algebra at tower level k , define $\mathcal{F}(i, j, l) = 1$ iff $\alpha(i, j, l) = \beta(i, j, l)$, where α and β are the left and right parenthesisisation signs from (1).

Theorem 3.4 (The 168 Theorem). *For $k = 3$ (octonions):*

1. $|\{(i, j, l) : \mathcal{F}(i, j, l) = 1\}| = 175$.
2. $|\{(i, j, l) : \mathcal{F}(i, j, l) = 0\}| = 168 = |\text{PSL}(2, 7)|$.
3. $175 + 168 = 343 = 7^3$.

Proof. Among the $7^3 = 343$ ordered triples, exactly $343 - 7 \cdot 6 \cdot 5 = 133$ have a repeated index; these are all associative by the alternativity of the octonions (any subalgebra generated by two elements is associative, by Artin’s theorem). Among the 210 triples with three distinct indices, each of the 7 Fano lines contributes $3! = 6$ ordered triples, giving 42 associative distinct triples. The remaining $210 - 42 = 168$ distinct triples are non-associative. Hence $175 = 133 + 42$ and $168 = 210 - 42$. This count coincides with the well-known order $|\text{PSL}(2, 7)| = 168$ of the automorphism group of $\text{PG}(2, 2)$.

Independently verified in Lean 4 via `native_decide (fano_count_175, non_fano_count_168` in `SounioCayleyDickson.lean`). $\square$

3.2 The Dagger Decomposition

Define the dagger on triples by reversal: $\dagger(i, j, l) = (l, j, i)$.

Theorem 3.5 (84/84 Duality). *The 168 non-Fano triples decompose under the dagger into:*

1. 84 forward triples with $\text{wave}(i, j, l) = +2$.

2. 84 backward triples with $\text{wave}(i, j, l) = -2$.

3. For every forward triple (i, j, l) , the dagger (l, j, i) is backward: $\text{wave}(l, j, i) = -\text{wave}(i, j, l)$.

4. No non-Fano triple is a fixed point of the dagger.

Proof. By exhaustive computation (theorems `arrow_forward_84`, `arrow_backward_84`, `dagger_reversal`, and `no_nonfano_self_dual` in `SounioCayleyDickson.lean`). $\square$

Corollary 3.6. *The 343 imaginary triples admit a tripartition:*

$$\underbrace{175}_{\text{invertible}} + \underbrace{84}_{\text{forward}} + \underbrace{84}_{\text{backward}} = 343$$

with the dagger preserving the associative set setwise (not pointwise: (i, j, l) and (l, j, i) are generally distinct ordered triples) and acting as a free involution on the 168 non-associative triples.

3.3 The Tamari Lattice and Selective Invertibility

The Tamari lattice T_n [22] is the partial order on binary trees with n leaves, where the covering relation is a single right-rotation: $((a \cdot b) \cdot c) \leq (a \cdot (b \cdot c))$.

In a Fano-selective structure, each edge of the Tamari lattice corresponding to a Fano triple becomes *undirected* (the rotation is invertible), while edges corresponding to non-Fano triples remain *directed*. The result is a partially directed Tamari lattice—a Hasse diagram where some edges are arrows and others are undirected.

The number of undirected edges is determined by the Fano predicate. For $n = 3$ factors there are $C_2 = 2$ parenthesisations and a single reassociation edge; this edge is undirected iff the triple is associative. The first nontrivial mixed picture appears at $n = 4$ factors ($C_3 = 5$ parenthesisations, 5 edges), where some edges are directed and others are not, depending on the Fano status of the constituent triples.

3.4 Skew Monoidal Coherence

We now prove that the Fano-selective associator structure carries a full skew monoidal coherence in the sense of Lack–Street [12]. The five axioms reduce to two verifiable conditions: the pentagon (3-cocycle) and the triangle (unitor) identities.

The reassociation factor. Define $\varphi(a, b, c) := \alpha(a, b, c) \cdot \beta(a, b, c)$, where α and β are the left and right parenthesisatation signs. Since $\alpha, \beta \in \{+1, -1\}$, we have $\varphi \in \{+1, -1\}$, with $\varphi = 1$ iff the triple is associative (Fano).

Unitors. The sign function satisfies $\sigma(0, a) = \sigma(a, 0) = 1$ for all $a \in \{0, \dots, 7\}$, so the identity element e_0 is both a left and right unit. The left and right unitors λ_a and ρ_a are identities. The four triangle axioms involving λ , ρ , and α are therefore trivially satisfied.

The pentagon. The pentagon axiom for a skew monoidal category requires, for all (a, b, c, d) :

$$\varphi(a \oplus b, c, d) \cdot \varphi(a, b, c \oplus d) = \varphi(a, b, c) \cdot \varphi(a, b \oplus c, d) \cdot \varphi(b, c, d) \quad (2)$$

This is exactly the *3-cocycle condition* on $\mathbb{Z}_2^3$ for the function φ . Albuquerque and Majid [4] showed that the Cayley–Dickson sign function determines a normalised 3-cocycle on $\mathbb{Z}_2^k$ for all k ; equation (2) is a direct consequence.

The ambient category. We work in the category $\mathcal{V}$ of $\mathbb{Z}_2^3$ -graded one-dimensional real vector spaces: objects are the 8 spaces $V_g = \mathbb{R} \cdot e_g$ ($g \in \mathbb{Z}_2^3$), tensor is induced by grade addition ($V_g \otimes V_h = V_{g \oplus h}$) with the Cayley–Dickson sign, unit is V_0 , and the associator on homogeneous elements is multiplication by $\varphi(g, h, k)$.

Theorem 3.7 (Cocycle Coherence). *The category $\mathcal{V}$ with tensor $\oplus$ (XOR), unit V_0 , and associator factor φ satisfies all five monoidal coherence axioms. In particular, φ is a normalised 3-cocycle on $\mathbb{Z}_2^3$, and the associator is an isomorphism on every triple.*

Proof. The unitor identities $\sigma(0, a) = \sigma(a, 0) = 1$ are verified for all 64 pairs (theorem `left_unitor_identity`, `right_unitor_identity` in `SounioSkewCategory.lean`). The pentagon (2) is verified for all $8^4 = 4096$ quadruples (theorem `pentagon_coherence`). The four triangle axioms follow from $\lambda = \rho = \text{id}$. Since $\varphi \in \{+1, -1\}$ and $(-1)^{-1} = -1$, every associator is invertible; $\mathcal{V}$ is a (non-symmetric) monoidal category with a $\mathbb{Z}_2^3$ cocycle twist, as in Albuquerque–Majid [4]. $\square$

3.5 The Rewrite Category

Remark 3.8. The algebraic associator is everywhere invertible, but this does *not* mean every reassociation is a valid compiler transformation. We now define the category in which the selective structure lives.

Definition 3.9 (Rewrite category). Let $\mathcal{R}$ be the category whose objects are fully parenthesised expressions over octonion basis elements, and whose morphisms are *semantics-preserving rewrites*: reassociation steps $(e_i \cdot e_j) \cdot e_l \rightarrow e_i \cdot (e_j \cdot e_l)$ that do not change the computed value. A morphism exists from the left parenthesisation to the right iff $\varphi(i, j, l) = +1$.

In $\mathcal{R}$, the 175 associative triples have morphisms (and their inverses, since the identity map is its own inverse); the 168 sign-changing triples have *no morphism at all*—not even a directed one. This is stronger than a skew monoidal category (which would have a directed morphism without an inverse); it is a *partial* monoidal structure where some reassociations are simply absent.

The `NonAssoc` effect in the compiler tracks exactly this distinction: the effect boundary marks the triples for which $\mathcal{R}$ has no morphism, preventing the e-graph from introducing rewrites that would change the program’s semantics.

4 Implementation in Sounio

Sounio is a self-hosted programming language for epistemic and scientific computing. Its compiler pipeline includes an algebraic effect system and an e-graph-based optimiser. We implement Fano-selective invertibility at two levels: the type checker and the optimiser.

4.1 The NonAssoc Algebraic Effect

We add `NonAssoc` as effect ID 14 in the effect system. Any function that multiplies non-associative hypercomplex values must declare this effect:

```
fn octonion_product(a: Hyper<Octonion, f64>,
                   b: Hyper<Octonion, f64>)
  -> Hyper<Octonion, f64> with NonAssoc {
  a * b
}
```

The type checker enforces the effect requirement:

```
// In check_hyper_binary (self-hosted/check/hyper.sio):
if !is_associative(result_alg) {
  if op == BinaryOp::OpMul {
    // E035: requires NonAssoc effect
  }
}
```

The effect triggers for all Cayley–Dickson algebras at tower level $k \geq 3$ (octonions, sedenions, and beyond).

4.2 The Fano Optimisation Predicate

The compiler’s reassociation optimiser uses the Fano predicate to decide safety:

```
// In ir_can_reassociate_triple (self-hosted/ir/algebra.sio):
fn ir_can_reassociate_triple(bits: i64, i: i64,
  j: i64, k: i64) -> bool with Mut, Panic, Div {
  if bits < 3 { return true }
  if i == 0 || j == 0 || k == 0 { return true }
  if i == j || j == k || i == k { return true }
  let alpha = ir_cd_sigma(i, j, bits)
    * ir_cd_sigma(i ^ j, k, bits)
  let beta = ir_cd_sigma(j, k, bits)
    * ir_cd_sigma(i, j ^ k, bits)
  alpha == beta // Fano line iff signs agree
}
```

The e-graph saturation engine checks this predicate before applying the associativity rewrite rule. When `algebra_kind` ≥ 3 (non-associative), the guard blocks the rewrite unless `ir_can_reassociate_triple` returns true.

This makes PG(2, 2) an operational component of the compiler: the 7 Fano lines determine the 42 distinct-index safe reassociations; together with the repeated-index cases (guaranteed by alternativity), these yield the 175 associative triples out of 343.

Remark 4.1 (Hard block vs. negation injection). An equality saturation engine could, in principle, apply the non-Fano reassociation and inject a compensating unary negation node into the e-class. We deliberately choose a *hard block* instead: in the category of semantic rewrites (Remark 3.8), a sign-changing transformation is not an identity but an effect boundary. Injecting arithmetic to compensate would silently introduce an operation that the programmer did not write and that the `NonAssoc` effect is designed to make visible. The hard block preserves the invariant that every rewrite in the e-graph is semantics-preserving *without* auxiliary compensation.

4.3 Artin’s Theorem as a Compiler Optimisation

For alternative algebras ($k = 3$, octonions), Artin’s theorem guarantees that any subalgebra generated by two elements is associative. The compiler implements this:

```
fn ir_artin_safe(alg: i64,
  n_distinct_vars: i64) -> bool {
  if alg <= 2 { return true }
  if alg == 3 && n_distinct_vars <= 2
  { return true }
  return false
}
```

This is a coarser but faster check: if an expression involves at most two distinct variables, the compiler can reassociate freely even in the octonions. Note that `n_distinct_vars` ≤ 2 captures

exactly the 133 repeated-index triples from the partition table in §2—the triples that are associative by alternativity (Artin’s theorem), independent of any Fano-line geometry. The Fano predicate provides finer-grained control for expressions with three or more distinct variables, recovering the remaining 42 associative triples from the 7 Fano lines.

5 Formal Verification in Lean 4

All results are formalised in Lean 4 [16] across four files totalling approximately 1,250 lines, with no uses of `sorry` and no dependency on Mathlib.

5.1 Decidability Strategy

The central observation enabling the formalisation is that every property of interest—the Fano predicate, the sign function, the associator coefficient, the 3-cocycle condition, the dagger decomposition—is a *decidable* predicate over a *finite* domain. For the octonions ($k = 3$), the imaginary basis indices range over $\{1, \dots, 7\}$; including the identity e_0 , the full basis is $\{0, \dots, 7\}$.

We exploit this by structuring every theorem as a Boolean computation verifiable by Lean’s `native_decide` tactic, which compiles the decision procedure to native code and executes it during elaboration. This gives us:

- **Exhaustive triple enumeration.** All $7^3 = 343$ imaginary triples are generated as a `List (Nat × Nat × Nat)` via `flatMap` over `List.range' 1 7`. Counting theorems (e.g., $|\mathcal{F}| = 175$) are stated as `(allImagTriples.filter p).length = 175` and closed by `native_decide`.
- **Exhaustive quadruple enumeration.** The pentagon coherence condition requires checking all $8^4 = 4096$ quadruples over $\{0, \dots, 7\}$. The enumeration is analogous.
- **No Mathlib.** The sign function uses only `Nat` arithmetic (`Nat.shiftLeft`, bitwise AND `&&&`, bitwise XOR `^^^`), all of which are kernel-computable in Lean 4. No algebra library, category theory library, or `Finset` machinery is needed. This keeps the trusted code base minimal and the build time under 3 seconds.
- **Termination.** The recursive sign function `cdSigma` terminates by structural decrease on the `bits` parameter, verified by `termination_by bits with decreasing_by omega`.

This approach trades generality for certainty: the theorems are proved for $k = 3$ (octonions) by exhaustive computation, not for arbitrary k by structural induction. Generalisation to higher Cayley–Dickson levels ($k \geq 4$) requires only increasing the enumeration range, at the cost of longer `native_decide` execution times ($15^4 = 50,625$ quadruples for $k = 4$).

5.2 Architecture

File	Theorems	Build time
<code>SounioCayleyDickson.lean</code>	13	666 ms
<code>SounioSkewCategory.lean</code>	18	672 ms
<code>SounioBidirectionalBridge.lean*</code>	18	420 ms
<code>SounioEffects.lean</code> (modified)	73	621 ms
Total	122	2.4 s

No sorry. No Mathlib dependency. All counting theorems are proved by `native_decide`, exploiting the decidability of Boolean predicates over finite domains. (`*SounioBidirectionalBridge.lean` proves that the α/β sign decomposition, when labelled as “synthesis/checking”, reproduces the same tripartition. This is a definitional relabelling, not an independent result; we include it for completeness and cross-validation.)

5.3 The Sign Function in Lean 4

The Lean formalisation mirrors the Sounio implementation:

```
def cdSigma (a b bits : Nat) : Int :=
  if a = 0 || b = 0 then 1
  else if bits <= 1 then -1
  else
    let half := Nat.shiftLeft 1 (bits - 1)
    let aHi := a >= half
    let bHi := b >= half
    let aLo := a &&& (half - 1)
    let bLo := b &&& (half - 1)
    if !aHi && !bHi then cdSigma aLo bLo (bits-1)
    else if !aHi && bHi then cdSigma bLo aLo (bits-1)
    else if aHi && !bHi then
      if bLo = 0 then cdSigma aLo bLo (bits-1)
      else -(cdSigma aLo bLo (bits - 1))
    else
      if bLo = 0 then -(cdSigma bLo aLo (bits-1))
      else cdSigma bLo aLo (bits - 1)
termination_by bits
```

5.4 Cross-Validation

The Lean formalisation and the Sounio implementation are cross-validated by three runtime verification programs (`examples/categorical_bridge_*.sio`) that independently compute all 343 triples and verify the same counts. The theorem `compiler_predicate_match` in Lean formally states that the Fano predicate agrees with the associator-coefficient-zero predicate for all 343 triples.

6 Related Work

Skew monoidal categories. Szlachányi [21] and Lack–Street [12] introduced skew monoidal categories to study bialgebroids. Bourke–Lack [6] connected them to skew multicategories. Buckley et al. [7] studied the Catalan simplicial set as a classifying structure for skew monoidal categories. Uustalu, Veltri, and Zeilberger [23] studied partially normal skew monoidal categories, where selected structural laws become invertible—the closest precedent to our work. However, their invertibility conditions are uniform (e.g., all left unitors are invertible), not indexed by a geometric predicate on basis triples.

Dagger categories and PL. Sabry et al. [19] implement daggers as a language feature in Π /Quantum Π . Karvonen [11] developed dagger category theory systematically. Heunen–Karvonen [10] studied dagger monads. None of these works connect daggers to non-associative algebra or the Fano plane.

Octonions as quasialgebra. Albuquerque–Majid [4] showed the octonions are an algebra object in a monoidal category with a non-trivial $\mathbb{Z}_2^3$ 3-cocycle associator. This resolves non-associativity by moving to a twisted monoidal category. Our approach preserves non-associativity as a first-class feature, selectively tracked by the effect system.

Non-associative Lambek calculus. Lambek [13] introduced the non-associative calculus NL with binary tree sequents. Buszkowski established polynomial-time decidability. A recent Agda formalisation [2] proves cut admissibility and interpolation. NL is a substructural logic, not a type system with algebraic effects; we share the binary tree structure but add the Fano selection and effect tracking.

Formal verification of algebra. Cowles–Gamboa [8] verified Cayley–Dickson properties in ACL2. Our work provides the first Lean 4 formalisation of the 168 theorem, the two-valued associator property, the dagger decomposition, and the pentagon coherence condition. The `native_decide` strategy for combinatorial algebraic identities may be of independent interest to the automated reasoning community.

Lean 4 formalisations of algebraic structures. The Lean 4 ecosystem includes Mathlib’s extensive algebraic hierarchy, but our formalisation deliberately avoids Mathlib to keep the trusted code base minimal and the build reproducible without external dependencies. The closest methodological precedent is the use of `decide/native_decide` for finite group theory computations in Lean, where exhaustive enumeration replaces structural induction for small instances.

7 Conclusion

We have introduced Fano-selective invertibility, a categorical structure in which the geometry of the Fano plane determines which associator morphisms are invertible. This structure:

- Generalises both monoidal and skew monoidal categories.
- Is instantiated by the Cayley–Dickson octonion algebra with the exact partition $175 + 84 + 84 = 343$.
- Is implemented as the `NonAssoc` algebraic effect and the Fano-based reassociation predicate in a working self-hosted compiler.
- Is formalised in Lean 4 with 122 machine-checked theorems.

Open questions.

1. Does Fano-selective invertibility extend to a full *Fano-selective skew closed category*, with non-invertible evaluation and currying maps for non-Fano triples?
2. For higher Cayley–Dickson algebras ($k \geq 4$), what is the structure of the Fano-selective partition? The ratio of non-Fano to total triples approaches $1/2$ as $k \rightarrow \infty$ (Corollary C of [3]).
3. The $84/84$ dagger decomposition exhibits the same combinatorial symmetry as the forward/backward states in time-symmetric quantum formalisms. Is this an accident of small numbers, or does it reflect a deeper connection through compact closed categories with daggers [1]?

Artifact. We freeze the self-hosted compiler artifact at release tag `self-host-native-v1` (commit `1aef4e65`). At this checkpoint the native compiler reaches a bit-identical fixed point (`gen2 == gen3`, MD5 `7448f33f5a5024fbada0c70a75e616ce`), passes 82/82 native runtime proofs, 16/16 native typecheck proofs, and 4/4 ARM64 compile proofs, with no Rust JIT required in the validation path. The Lean 4 formalisation, Sounio compiler, and runtime verification examples are available at <https://github.com/Sounio-lang/sounio>.

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